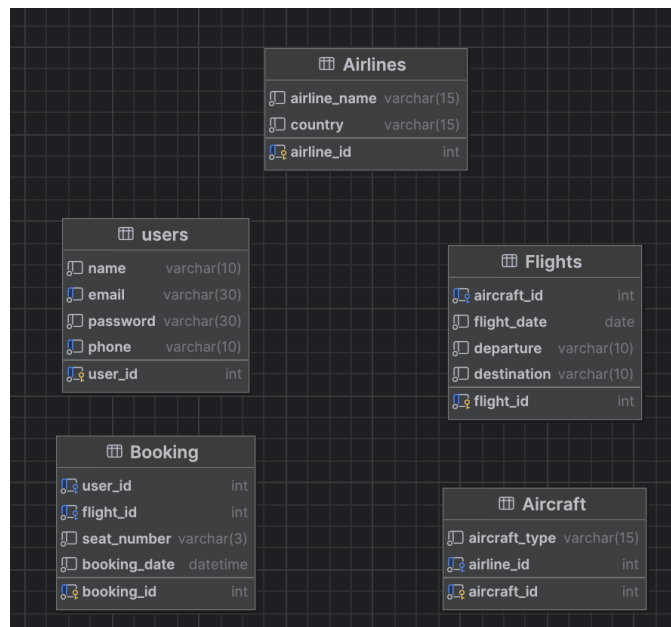


Airline System

A database system for managing **users, airlines, aircraft, flights, and flight bookings**. It allows users to book flights and stores the related flight, aircraft, airline, seat, and booking information.



This diagram shows the database tables and their attributes, including the data types, primary keys, and foreign keys for each table.

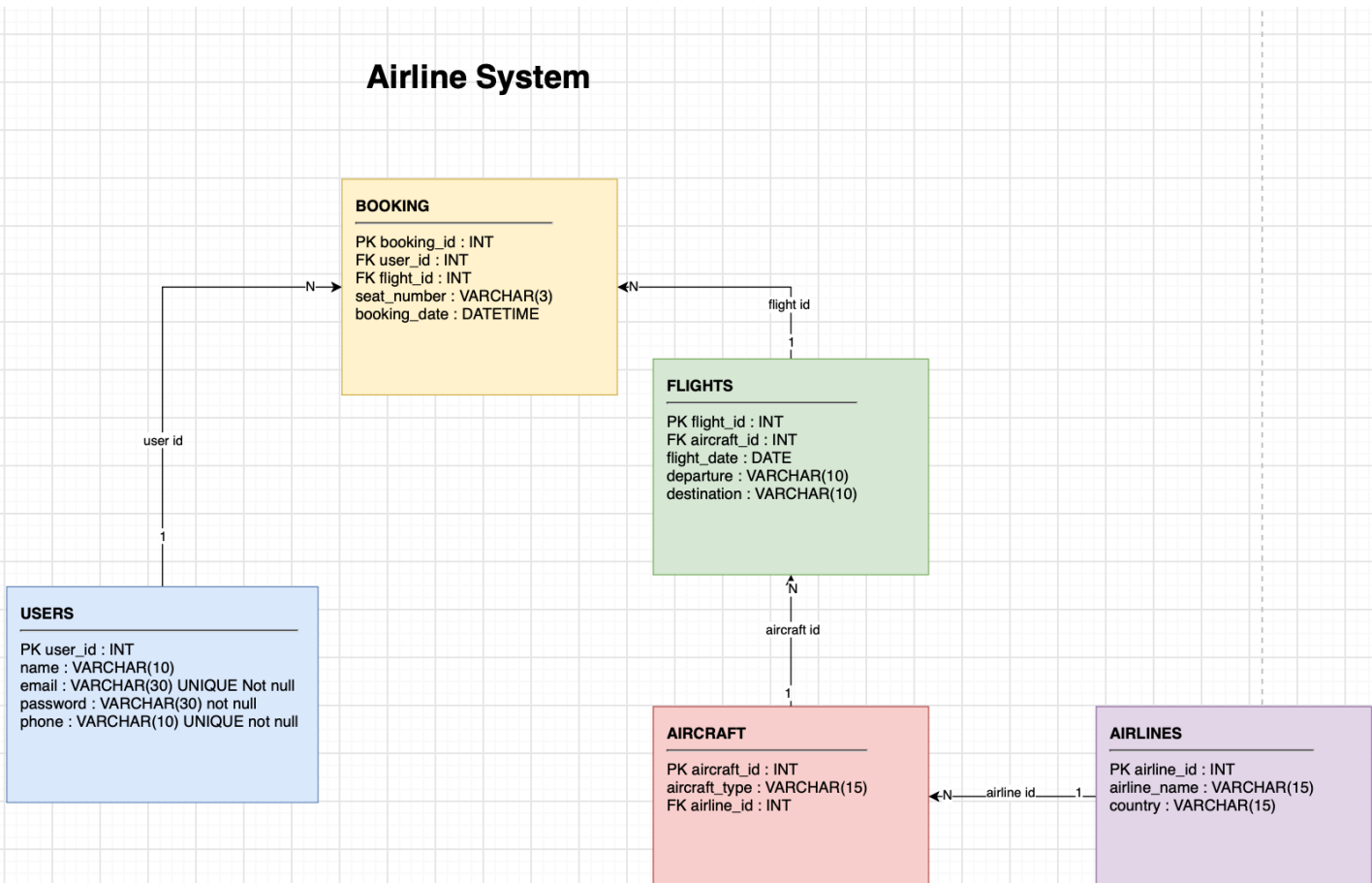
Database Schema – Airline System

This ER diagram represents the database structure of the Airline System and shows the relationships between the five main tables:

- **Users → Booking:** One user can have many bookings (**1:N**).
- **Flights → Booking:** One flight can have many bookings (**1:N**).
- **Aircraft → Flights:** One aircraft can be used for many flights (**1:N**).
- **Airlines → Aircraft:** One airline can have many aircraft (**1:N**).

The diagram also shows the **Primary Keys (PK)** and **Foreign Keys (FK)** used to connect the tables and maintain the relationships between them.

[View the ER Diagram](#)



```
create database Airline_System;

use Airline_System;
create table users(
    user_id int primary key ,
    name varchar(10) not null,
    email varchar(30)not null unique ,
    password varchar(30) not null ,
    phone varchar(10) not null unique
);

create table Airlines(
    airline_id int primary key ,
    airline_name varchar(15) not null,
    country varchar(15)not null
);

create table Aircraft(
    aircraft_id int primary key ,
    aircraft_type varchar(15) not null ,
    airline_id int not null ,
    foreign key (airline_id)references Airlines(airline_id)
);

create table Flights (
    flight_id int primary key ,
    aircraft_id int not null ,
    foreign key (aircraft_id)references Aircraft(aircraft_id),
    flight_date DATE not null ,
    departure varchar(10)not null ,
    destination varchar(10)not null
);

create table Booking(
    booking_id int primary key ,
    user_id int not null,
    foreign key (user_id)references users(user_id),
    flight_id int not null ,
    foreign key (flight_id)references Flights(flight_id),
    seat_number varchar(3)not null ,
    booking_date datetime not null
);
```

```
);

insert into users values (111, 'ali', 'ali@gmail.com', 'ali111222', '0540577988')
, (112, 'shumua', 'shumua@gmail.com', 'shumua1122', '0537466755');

insert into Airlines values (221, 'Saudia', 'KSA'),
(222, 'flynas', 'KSA');

insert into Aircraft values (331, 'Boeing 777', 221),
(332, 'Airbus A320', 222);

insert into Flights values (441, 331, '2026-09-20', 'jazan', 'Jeddah'),
(442, 332, '2026-09-21', 'Riyadh', 'Dubai');

insert into Booking values (001, 111, 441, 'A01', NOW()),
(002, 112, 442, 'A40', NOW());
```